

DLS Buddy — Quickstart Guide

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1. Introduction

DLS Buddy analyzes **dynamic (DLS)** and **static (SLS)** light scattering data. It is not tied to any instrument, solvent, or polymer: you supply every physical parameter — temperature, viscosity, refractive index, wavelength, scattering angle, and (for SLS) dn/dc and concentration — and nothing is assumed or hard-coded.

This guide covers the common path — load a file, confirm its parameters, run a DLS or SLS analysis, and export the result. For every supported format, the Cross-Sample and Utilities tools, and the full parameter reference, see the **User-Manual**. For the equations and the literature behind each method, see the **Theory-and-Equations-Guide**.

The window is a left **sidebar** (your loaded data, auto-grouped into samples) plus tabbed modules. The two you need first are **Data** (confirm parameters) and then **DLS** or **SLS** (analyze).

Launching the program

Double-click the launcher for your operating system:

- **Windows** — Launch DLS Buddy (Windows).bat
- **macOS** — Launch DLS Buddy (MacOS).command (on the first run, macOS Gatekeeper may refuse an unsigned script — right-click it → **Open** → confirm once, and it opens normally thereafter)

Or run it from the repository root:

```
python -m gui.main
```

2. The Workflow at a Glance

Almost every analysis follows the same short path:

Launch → **Load** → **Confirm parameters** + **Update** → **open DLS or SLS** → **pick a method and Run** → **read the result** → **Export**.

1. **Launch** the program.
2. **Load** your data — **Load DLS correlogram...** or **Load SLS intensities...**. Files are auto-detected and grouped into **samples** in the sidebar (§3).
3. **Confirm the parameters** on the **Data** tab, then click **Update** to commit — analysis runs on the committed values (§4).
4. **Open the analysis tab:** **DLS** for size and R_h (§5), **SLS** for M_w , R_g , and A_2 (§6).
5. **Pick a method and Run**, then read the result in the table and plot. Flags mark *apparent* vs *thermodynamic* and *uncalibrated* results so they aren't over-read.
6. **Export** with **Export CSV...** (§7).

3. Loading Data

Click **Load DLS correlogram...** or **Load SLS intensities...** and pick one or more files. You do **not** choose an instrument — the program **auto-detects** the format, trying each supported instrument parser (Brookhaven, Malvern Zetasizer, ALV) until one recognizes the file, and falling back to a plain two-column text file if none do. Any file that can't be read is reported at the end; the rest still load.

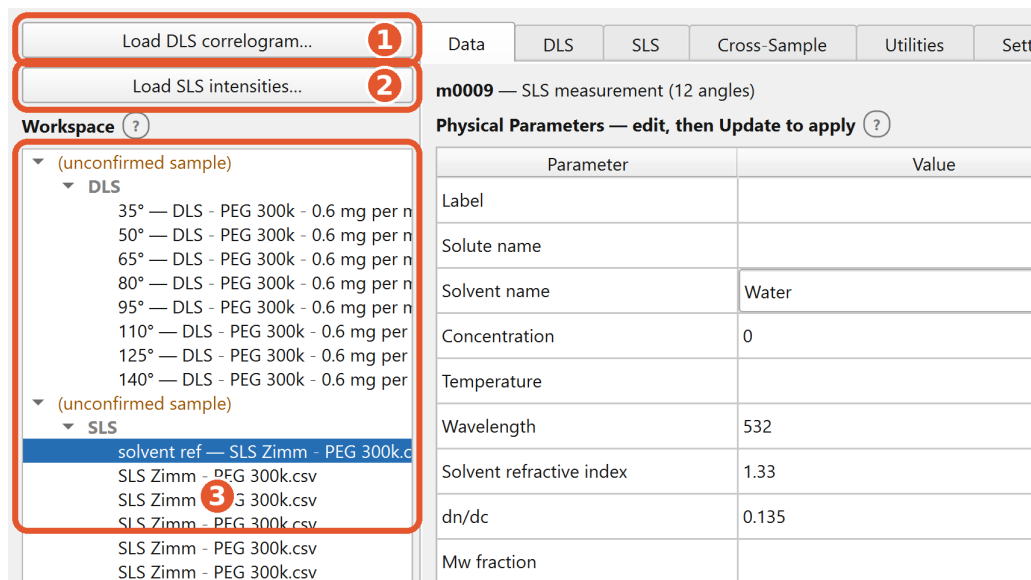


Figure 1: Loading data: click **Load DLS correlogram...** (1) or **Load SLS intensities...** (2). The loaded measurements appear in the **Workspace** sidebar (3), auto-grouped into a sample — here one file has loaded eight scattering angles — and their parameters populate the **Data** tab, ready to confirm (§4).

The parser pre-fills whatever parameters the file contains. Pre-filled values are suggestions, not

locked inputs — you can override any of them on the **Data** tab. If a file is in none of the supported formats, the program says so rather than guessing.

Note. Some formats carry no metadata (a plain-text correlogram, a Zetasizer clipboard record), so you enter every parameter at the confirmation step. For the full list of formats and exactly what each one pre-fills, see the **User-Manual**.

4. Confirming Parameters (Data tab)

The **Data** tab shows every parameter for the selected measurement. Fill in any blank field and correct any pre-filled value, then click **Update** to commit.

Physical Parameters — edit, then Update to apply ?

Parameter	Value	Unit
Label		
Solute name		
Solvent name		
Concentration		mg/mL
Temperature	25	°C
Scattering angle	35	°
Wavelength	532	nm
Solvent refractive index	1.33	
Viscosity	0.89	mPa·s
Mw fraction		
Polarization (DDLS)	—	

• library value (auto-filled) · no dot = your value · a value you type is never overwritten

Update Undo Reset

Figure 2: On the **Data** tab, edit each parameter and choose its unit in the dropdown (1), then press **Update** (2) to commit. A green dot marks a value the solvent library auto-filled; a value you type yourself is never overwritten.

- **Identity and optics are entered once per sample** — polymer/sample name, solvent, temperature, refractive index, wavelength, and scattering angle. Press **Enter** to jump to the next field.

- Each parameter has a **Unit** column: a dropdown where a quantity has several natural units (switching units re-displays the value; the stored physical value is unchanged), or a fixed label otherwise. Defaults are on a human scale (°C, mPa·s, mg/mL, nm).
- For common solvents, picking the **Solvent name** from the dropdown fills in the refractive index and viscosity from the sample's temperature and wavelength, each marked with a small **teal dot**. A value you type by hand is never overwritten and shows no dot. **dn/dc is never proposed** — it is the most sensitive input for low-contrast systems, so you always enter it yourself.
- **Concentration** is required for SLS and for the DLS concentration series; single cumulant / exponential / distribution size analysis does not use it.

Analysis always runs on the **committed** values. If you change a parameter that fed a fit already on screen, the result is kept but flagged stale until you re-run it. The full unit table and the solvent library are in the **User-Manual**.

5. DLS — the Common Path

The **DLS** tab extracts particle dynamics — decay rates, diffusion coefficients, and hydrodynamic radii — from a measured $g_2(\tau)-1$. For a quick size, you need two of its sub-tabs.

Tick the measurement(s) to analyze in the **Measurements to plot** checklist. This checklist is the sole driver of what gets fit; selecting a measurement in the sidebar only navigates. The raw correlogram appears as soon as you tick it.

Then choose one route:

- **Correlogram** — parametric fits. The **cumulant** fit gives the intensity-weighted (z-average) R_h and the polydispersity index (PDI); single/double exponential and KWW fits are also here. Set the cumulant order in this sub-tab. A PDI above 0.3 is flagged as unreliable.
- **Distribution** — **CONTIN** / **NNLS** / **Lognormal**. These recover a full distribution of sizes without assuming a shape: NNLS resolves well-separated peaks but is spiky, CONTIN is smoother and generally better for broad or real data, and Lognormal is a robust single-mode default. Resolved peaks are labelled on the plot and listed in the **Peak results** table.

Set the **delay window** and **baseline region** on the Correlogram sub-tab — type the values or drag the markers on the plot (green carets along the top are the window; grey carets along the bottom are the baseline). They apply to every ticked measurement.

Click **Run** to fit, and read the result in the table and plot. A single cumulant or distribution peak is an **apparent** R_h at one angle and concentration.

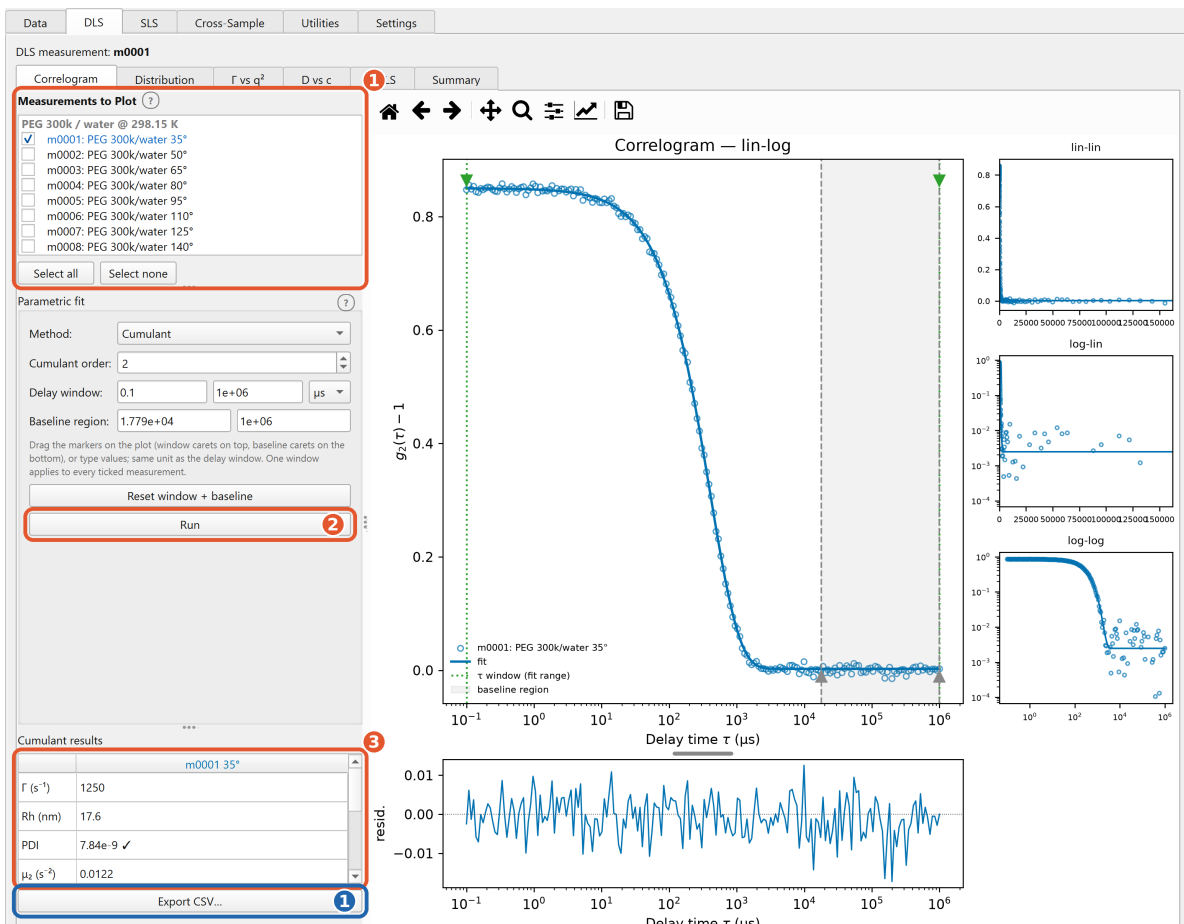


Figure 3: The DLS **Correlogram** sub-tab. *Orange*: tick the measurement(s) to plot (1) and press **Run** (2); read R_h and PDI in the results table (3). The correlogram and its cumulant fit fill the plot on the right. *Blue*: **Export CSV...** writes the result on screen (§7).

Tip. Fitting many heavy distributions (especially CONTIN) at once is slow — tick only what you want to compare.

The multi-angle (Γ vs q^2), concentration-series (**D vs c**), and depolarized (**DDLS**) analyses, and the workspace-wide **Summary**, are covered in the User-Manual.

6. SLS — the Common Path

The **SLS** tab extracts the weight-average molecular weight M_w , the radius of gyration R_g , and the second virial coefficient A_2 . Pick the sample from the **SLS sample** dropdown at the top; the whole set — solvent reference plus the concentration series — is analyzed together.

1. **Set the calibration.** The program computes its own calibration constant kc from one calibrant measurement — the intensity of a standard liquid (usually toluene), the angle, and that standard's geometry-aware Rayleigh factor. Any value stored in the instrument file is kept for reference only and never used. If you supply no calibration, the analysis still runs on an arbitrary scale, flagged **uncalibrated**: then M_w and the absolute A_2 are unreliable, but

- R_g and the calibration-free product $2A_2Mw$ remain valid.
2. **Choose a method.** **Zimm** or **Berry** double extrapolation combines several concentrations and angles, extrapolates to zero angle and zero concentration, and returns Mw , R_g , and A_2 at once — a **thermodynamic** result. (Berry linearizes the high- qR_g regime of large Mw better than Zimm; the program reports which it used.) A single-concentration **Debye** fit gives only **apparent** values.
 3. **Mask any bad points** (click to hide an angle, a concentration, or a single point), then click **Run** and read the result. Flags mark apparent vs thermodynamic and calibrated vs uncalibrated.

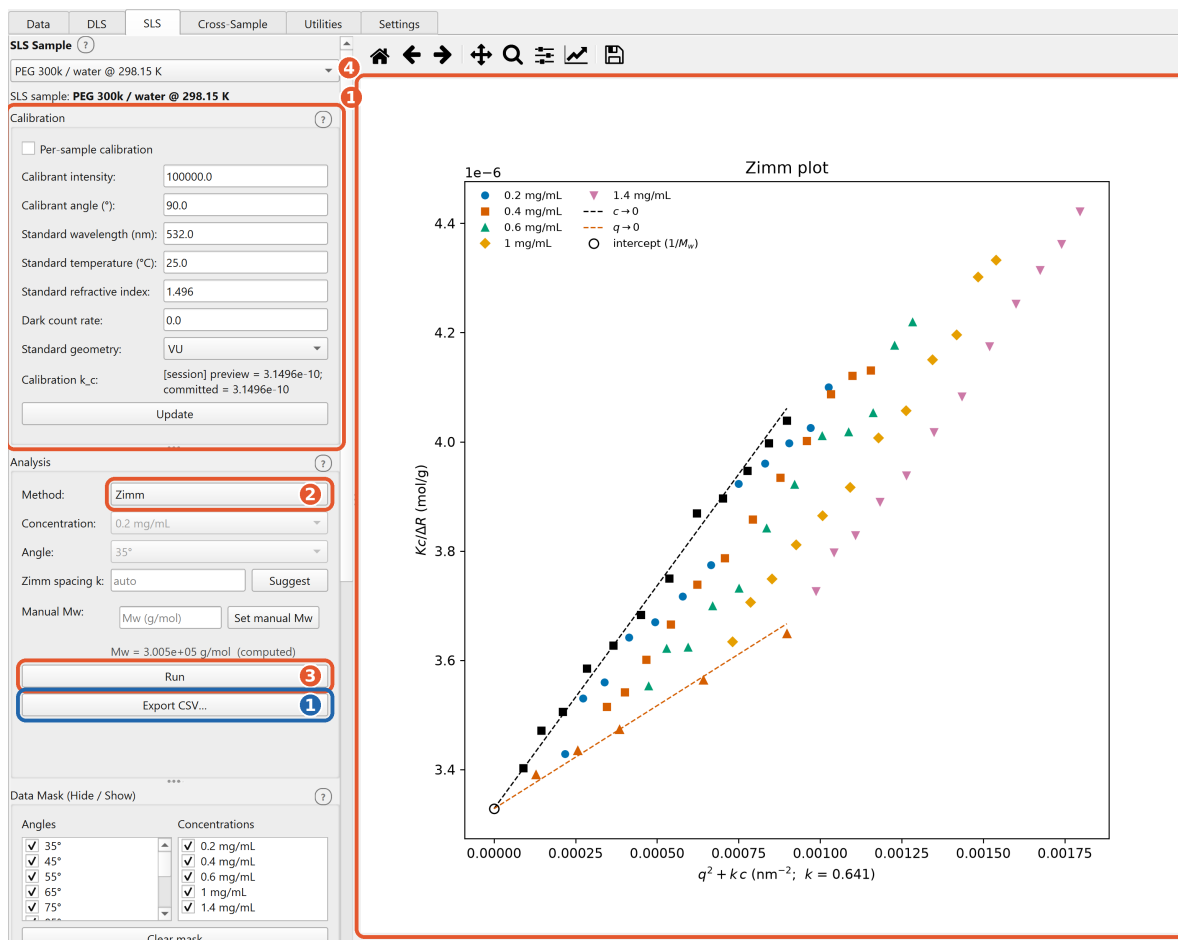


Figure 4: The **SLS** tab. *Orange*: set the calibration (1), choose the method — **Zimm** here (2) — and press **Run** (3); the Zimm double-extrapolation plot and Mw / R_g / A_2 appear (4). *Blue*: **Export CSV...** writes the result (§7).

Guinier, single-angle Mw , the calibration-free A_2 route, the depolarization correction, and the manual- Mw override are covered in the User-Manual.

7. Exporting Results

Every analysis view has an **Export CSV...** button that writes the result currently on screen to a CSV that OriginLab imports directly. Each file carries the four Origin label rows (Long Name, Units, Comments, Parameters), so axis labels and units populate on import. An uncalibrated result writes “uncalibrated, arbitrary scale” into the **Comments** cell of the affected columns only, without changing the row count, so the Origin import stays intact.

Data-quality flags are interface overlays, not part of the figure — a saved plot image is always clean.